Notes: Hansen & Jagannathan (JF, 1997)

Assessing Specification Errors in Stochastic Discount Factor Models

Contents

1 Big picture

- **Question.** How should we compare asset-pricing models when everyone knows that their stochastic discount factor proxies are misspecified and therefore do not price all test assets exactly?
- **Problem with classical tests.** A χ^2 test asks whether all pricing errors are zero. In many asset-pricing settings this null is implausible. Worse, a χ^2 criterion can reward highly variable discount-factor proxies because the weighting matrix changes with the proxy.
- **Main idea.** Measure misspecification by the least-squares distance between the candidate stochastic discount factor proxy and the set of stochastic discount factors that price the test assets correctly.
- **Two distances.** The paper defines δ, the distance to the set $\mathcal{M}$ of all valid SDFs that price the assets, and δ^+, the distance to the smaller set $\mathcal{M}^+$ of nonnegative SDFs.
- **Pricing-error interpretation.** δ is also the maximum pricing error over portfolios, normalized by the payoff's second moment. δ^+ is the corresponding maximum pricing error when one also uses the implications of nonnegative pricing for derivative payoffs.
- **Why positivity matters.** A discount factor that prices currently observed securities can still be negative in some states. Such an SDF would assign negative prices to some positive derivative payoffs. δ^+ asks how close the proxy is to an arbitrage-free pricing kernel.
- **Empirical application.** The paper applies the distance measures to power utility SDFs, a consumption-externality SDF, the reciprocal market-return proxy, and linear factor models.
- **Main empirical result.** Power utility SDFs have distance about 0.327–0.334 from the admissible SDF set. The reciprocal market-return proxy improves slightly to about 0.311. Simple linear factor models do better, with distances around 0.286–0.325.
- **Main methodological warning.** The model chosen by minimizing the χ^2 statistic need not be the model with the smallest economically meaningful misspecification. In Table IV, a consumption-growth factor can minimize χ^2 while producing a much larger specification error.
- **Takeaway.** For comparing knowingly misspecified asset-pricing models, distance-based specification errors are more informative than asking whether an exact pricing null can be rejected.

2 Data and measurement

2.1 Payoffs, prices, and stochastic discount factors

- Let $\mathcal{P}$ be the payoff space of portfolios or contingent claims.
- Let $\pi(p)$ be the price assigned to payoff $p \in \mathcal{P}$.
- A stochastic discount factor m prices payoffs correctly if

$$\mathbb{E}[mp] = \pi(p) \quad \text{for all } p \in \mathcal{P}. \tag{1}$$

- The admissible set is

$$\mathcal{M} = \{m \in L^2 : \mathbb{E}[mp] = \pi(p) \text{ for all } p \in \mathcal{P}\}. \tag{2}$$

- The arbitrage-free admissible set is

$$\mathcal{M}^+ = \{m \in \mathcal{M} : m \geq 0 \text{ with probability one}\}. \tag{3}$$

Interpretation: A model is represented by a candidate random variable y. The paper does not ask whether y is exactly correct. It asks how far y is from the set of SDFs that would be correct for the assets under study.

2.2 Distance measures

The first specification-error measure is

$$\delta(y) = \min_{m \in \mathcal{M}} \left(\mathbb{E}[(y-m)^2]\right)^{1/2}. \tag{4}$$

The second measure imposes nonnegativity:

$$\delta^+(y) = \min_{m \in \mathcal{M}^+} \left(\mathbb{E}[(y-m)^2]\right)^{1/2}. \tag{5}$$

Interpretation: δ measures how much the proxy must be adjusted, in least-squares terms, to price the assets. δ^+ asks for the smallest adjustment that also rules out negative pricing kernels.

2.3 Observable return data

The empirical section uses monthly returns from 1959:1 to 1990:12. The six test assets are:

- an equally weighted portfolio of NYSE stocks in the largest size decile;
- an equally weighted portfolio of NYSE stocks in the smallest size decile;
- a portfolio of long-term government bonds;
- three managed portfolios formed by putting $1 - z$ in the one-month Treasury bill and z in the largest size-decile stock portfolio.

The managed-portfolio weights z are lagged yield spreads:

- Aaa minus Baa bond yields;
- Baa bond yield minus the one-month Treasury bill yield;
- one-year Treasury bill yield minus one-month Treasury bill yield.

Interpretation: The managed portfolios bring conditioning information into an unconditional SDF test. They help reveal whether candidate SDFs miss predictable variation in returns.

2.4 Candidate stochastic discount factor proxies

The paper studies several proxies:

- **Constant SDF:**

  $$y_{t+1} = \beta. \tag{6}$$

  This is a risk-neutral benchmark.

- **Power utility SDF:**

  $$y_{t+1} = \beta \left(\frac{c_{t+1}}{c_t}\right)^{-\gamma}. \tag{7}$$

- **Power utility with consumption externality:**

  $$y_{t+1} = \beta (c_{t+1}/c_t)^{-\gamma} (c_t/c^*_{t-1})^{\gamma-1}, \tag{8}$$

  with $c^*_{t-1} = c_{t-1}$ in the empirical implementation.

- **Reciprocal market return:**

  $$y_{t+1} = \frac{1}{R_{m,t+1}}. \tag{9}$$

- **Observable linear factor models:**
$$y_{t+1} = a' f_{t+1}, \tag{10}$$
where f_{t+1} contains a constant and one variable factor: consumption growth, equal-weighted market return, or value-weighted market return.

Interpretation: The methods can compare both utility-based SDFs and factor-pricing SDFs on the same distance scale.

2.5 Sampling error and classical test statistics

- The paper reports standard errors for δ and δ^+ using Newey–West corrections with lag length $m = 15$.
- The paper also reports χ^2 statistics for the null hypothesis that all pricing errors are zero.
- The distance measures are meant for model comparison under misspecification; the χ^2 statistics are meant for exact specification tests.

Interpretation: The key distinction is practical. A very large χ^2 statistic says the model is rejected; δ tells how badly the proxy prices the asset menu.

3 Models and empirical specifications

3.1 Maximum pricing error interpretation

For a candidate proxy y, define the pricing error on payoff p as
$$e_y(p) = \mathbb{E}[yp] - \pi(p). \tag{11}$$
The paper shows that
$$\delta(y) = \max_{p \in \mathcal{P},\ \|p\|=1} |e_y(p)|, \tag{12}$$
where $\|p\| = (\mathbb{E}[p^2])^{1/2}$.

Interpretation: The distance is not an abstract geometric number. It is the largest pricing error the proxy makes on a portfolio whose payoff has unit second moment.

3.2 Finite-dimensional return version

Let R be a vector of gross returns with price vector $\mathbf{1}$. The pricing-error vector is
$$g(y) = \mathbb{E}[yR] - \mathbf{1}. \tag{13}$$
When the second-moment matrix $\mathbb{E}[RR']$ is nonsingular,
$$\delta(y)^2 = g(y)' \left(\mathbb{E}[RR']\right)^{-1} g(y). \tag{14}$$

Interpretation: This is the Hansen–Jagannathan distance formula. It standardizes pricing errors by the payoff covariance structure, so it measures the error of the most mispriced portfolio rather than treating each asset error independently.

3.3 Closest admissible discount factor

The closest valid SDF can be written as
$$m^*(y) = y - p_y, \tag{15}$$
where p_y is the smallest least-squares correction that removes the pricing errors.

Interpretation: The correction p_y identifies the payoff direction that the proxy misprices most. A large δ means a large correction is required to turn the proxy into a valid SDF.

3.4 Nonnegative SDF problem

The nonnegative distance is

$$\delta^+(y) = \min_{m \in \mathcal{M},\ m \geq 0} \|y - m\|. \tag{16}$$

The paper solves this using a conjugate optimization problem. For payoff vector x and price vector q, the closest nonnegative discount factor has the form

$$m^+(y, \lambda) = (y - \lambda' x)^+, \tag{17}$$

where $(z)^+ = \max\{z, 0\}$.

Interpretation: Nonnegativity matters because a negative SDF can price the observed asset span but still imply impossible prices for derivative claims with nonnegative payoffs.

3.5 Why the χ^2 statistic can mislead model comparison

A χ^2 statistic is based on a quadratic form in sample moments. The weighting matrix depends on the sampling variability of the moments. For misspecified SDF proxies, this can favor a proxy with high variability because the model's own variability changes the denominator of the statistic.

Interpretation: The distance criterion asks: how much must the SDF proxy be moved to price the assets? The χ^2 criterion asks: how surprising are the sample errors under an exact null? These are different questions.

3.6 Linear factor model connection

For a linear factor model, the SDF proxy is

$$y = a' f. \tag{18}$$

When the factor prices are chosen by least squares, the Hansen–Jagannathan distance links the implied SDF to the maximum expected payoff error in the associated factor-pricing model.

Interpretation: The paper connects SDF misspecification to factor-pricing errors. This makes the distance measure relevant for comparing CAPM-style and consumption-factor models.

4 Conceptual design and credibility of the method

4.1 What the method identifies

- The method identifies the smallest least-squares adjustment needed to make a proxy price the chosen asset menu.
- The adjustment depends on the assets included in the test. A model can have small distance for one asset menu and larger distance for another.
- The maximum-error portfolio provides an economic interpretation of where the proxy fails.

Interpretation: The paper does not claim a universal model ranking independent of assets. It provides a coherent ranking conditional on a chosen payoff space.

4.2 Why the payoff normalization matters

- Raw pricing errors cannot be compared across portfolios without controlling for payoff scale.
- The paper normalizes by the payoff second moment, $\|p\| = (\mathbb{E}[p^2])^{1/2}$.
- This makes the error criterion invariant to multiplying the payoff by a constant.

Interpretation: Normalization is what turns a vector of pricing errors into a single economically meaningful model-comparison statistic.

4.3 Why positivity is a separate issue

- If only the observed asset space is considered, many SDFs may price the data.
- Some of these SDFs may be negative in some states.
- Negative SDFs violate arbitrage-free pricing of derivative claims.
- δ^+ asks whether the model is close to a genuinely arbitrage-free SDF, not merely a linear functional on the observed assets.

Interpretation: In the empirical application, δ and δ^+ are almost identical, suggesting that the positivity constraint is not the main reason these proxies fail.

4.4 What the method cannot fully prove

- A small distance does not prove that a model is true; it only means the proxy prices the chosen assets comparatively well.
- The ranking can change if the asset menu changes, especially if more informative managed portfolios or derivatives are added.
- Sampling error remains important, especially with highly persistent or managed returns.
- The empirical application uses a limited set of six assets, so it should be read as a demonstration of the method rather than a final ranking of all asset-pricing models.

Interpretation: The contribution is methodological: it gives a disciplined way to compare misspecified SDFs without treating exact correctness as the maintained hypothesis.

5 Tables and figures

The paper has no figures. The original empirical tables are grouped below to save space and improve comparison; all table images are directly cropped from the uploaded paper.

5.1 Appendix Tables B.1–B.2: data summary

Read:

- Table B.1 reports means, standard deviations, skewness, and kurtosis for the six return series.
- Security 5 has the largest standard deviation, 0.167, and high kurtosis, 10.824.
- Table B.2 shows high return correlations among several securities, especially securities 4 and 5 at 0.93.
- The managed portfolios are highly correlated with equity-related returns, which affects the most-mispriced portfolio directions.

Economic message: The data are not just static benchmark assets. The managed portfolios embed conditioning information and make the SDF tests more demanding.

Summary Statistics of Return Data

The six securities are described in Section VI.A.

Security	Mean	Standard Deviation	Skewness	Kurtosis
1	1.008	0.044	−0.122	4.975
2	1.013	0.071	0.899	10.402
3	1.005	0.029	0.789	6.216
4	1.006	0.056	1.066	9.124
5	1.021	0.167	0.697	10.824
6	1.001	0.056	1.043	13.598

Table B.2

Correlations Among Security Returns

The six securities are described in Section VI.A.

Security	2	3	4	5	6
1	0.70	0.33	0.92	0.90	0.72
2		0.14	0.63	0.60	0.42
3			0.40	0.31	0.30
4				0.93	0.80
5					0.81

Table 1: Appendix Tables B.1–B.2: summary statistics and correlations of return data.

5.2 Tables I and III: power utility, with and without externality

Read:

- Table I shows that the power utility model has δ around 0.328–0.334 across most parameter values.
- The minimized power utility specification error is 0.327, with standard error 0.053.
- δ and δ^+ are essentially the same, so the positivity restriction barely changes the distance.
- The χ^2 statistic varies enormously across preference parameters, from thousands down to 35.2 at the minimized configuration.
- Table III adds consumption externality. The distance remains about 0.328–0.332 and the minimized δ is 0.328.
- The externality reduces the minimized χ^2 statistic to 27.5, but it does not materially reduce the distance measure.

Economic message: The utility specifications are rejected by classical tests, but the HJ distance gives a stable economic measure of misspecification. The externality changes the test statistic more than it changes the pricing-error distance.

Table I

Specification Errors for Power Utility

The stochastic discount factor proxy is $\beta(c_{t+1}/c_t)^{-\gamma}$, where β is the rate of time preference, c_t is the per capita nondurables consumption for month t, and γ is the coefficient of relative risk aversion of the representative consumer. Details of the data construction and sources are given in Section VI.A and Appendix B. The column labeled δ gives the least squares distances between the proxies and the family of stochastic discount factors, and the column labeled δ^+ gives the distances between the proxies and the family of nonnegative stochastic discount factors. The column labeled χ^2 gives the chi-square test statistics for the null hypothesis that all pricing errors are zero. Numbers in parentheses are estimated standard errors. The standard errors are computed using the method described in Newey and West (1987) with $m = 15$. (Very similar results are obtained using $m = 9$ and 12.) The entries in the row labeled "minimized" are computed by selecting parameter values for β and γ that minimized the corresponding criteria. In the column labeled δ the minimizers are $\beta = 1.03$ and $\gamma = 33.3$; in the column δ^+ the minimizers are $\beta = 1.03$ and $\gamma = 28.6$; and in the column labelled χ^2 the minimizers are $\beta = 1.608$ and $\gamma = 249.5$.

γ	δ	δ^+	χ^2
	$\beta = 0.95$		
0	0.332	0.333	2,824.9
	(0.054)	(0.054)	
1	0.333	0.333	2,850.9
	(0.053)	(0.054)	
5	0.333	0.333	1,557.7
	(0.053)	(0.054)	
10	0.333	0.333	705.9
	(0.053)	(0.053)	
15	0.334	0.334	419.2
	(0.052)	(0.053)	
	$\beta = 1.00$		
0	0.329	0.329	46.6
	(0.054)	(0.055)	
1	0.329	0.329	42.6
	(0.054)	(0.054)	
5	0.329	0.329	47.8
	(0.054)	(0.054)	
10	0.328	0.328	55.7
	(0.053)	(0.054)	
15	0.328	0.328	58.1
	(0.053)	(0.054)	
Minimized	0.327	0.327	35.2

Table 2: Paper Table I: power utility specification errors.

Table III

Specification Errors for Power Utility With Consumption Externality

The stochastic discount factor proxy is $\beta(c_{t+1}/c_t)^{-\gamma}(c_t/c_{t-1})^{\gamma-1}$, where β is the rate of time preference, c_t is the per capita nondurables consumption for month t, and γ is the coefficient of relative risk aversion of the representative consumer. Details of the data construction and sources are given in Section VI.A and Appendix B. The column labeled δ gives the least squares distances between the proxies and the family of stochastic discount factors, and the column labeled δ^+ gives the distances between the proxies and the family of nonnegative stochastic discount factors. The column labeled χ^2 gives the chi-square test statistics for the null hypothesis that all pricing errors are zero. Numbers in parentheses are estimated standard errors. The standard errors are computed using the method described in Newey and West (1987) with $m = 15$. The entries in the row labeled "minimized" are computed by selecting parameter values for β and γ that minimize the corresponding criteria. In the column labeled δ the minimizers are $\beta = 0.998$ and $\gamma = 7.3$; in the column δ^+ the minimizers are $\beta = 0.999$ and $\gamma = 4.8$; and in the column labelled χ^2 the minimizers are $\beta = 21.1$ and $\gamma = 237.6$.

γ	δ	δ^+	χ^2
	$\beta = 0.95$		
0	0.332	0.332	2,762.1
	(0.054)	(0.054)	
1	0.332	0.332	2,837.7
	(0.054)	(0.054)	
5	0.332	0.332	807.6
	(0.053)	(0.054)	
10	0.332	0.332	244.0
	(0.053)	(0.053)	
15	0.332	0.332	123.9
	(0.052)	(0.053)	
	$\beta = 1.00$		
0	0.328	0.339	42.2
	(0.054)	(0.055)	
1	0.328	0.328	42.3
	(0.054)	(0.055)	
5	0.328	0.328	42.1
	(0.054)	(0.055)	
10	0.328	0.329	42.8
	(0.053)	(0.054)	
15	0.328	0.329	42.8
	(0.053)	(0.053)	
Minimized	0.328	0.328	27.5
	(0.053)	(0.054)	

where we have imposed the equilibrium condition that $c^*_{t-1} = c_{t-1}$. The stochastic discount factor proxy is now

Table 3: Paper Table III: power utility with consumption externality.

5.3 Table II: Lagrange multipliers for power utility

Read:

- Table II reports Lagrange multipliers for the pricing constraints when the power utility proxy is corrected in the least-squares sense.
- The multipliers are very similar across $\gamma = 0$, 5, and 15.
- Security 4 has a large negative multiplier, around -13.5.
- Securities 5 and 6 also have economically important multipliers.
- The similarity of multipliers across parameter values explains why the δ distances in Table I change little across γ.

Economic message: The same payoff directions remain mispriced across many versions of the power utility proxy. Changing risk aversion does not fix the core pricing-error direction.

Table II

Lagrange Multipliers for Power Utility

The stochastic discount factor proxy is $\beta(c_{t+1}/c_t)^{-\gamma}$. The numbers reported are the Lagrange multipliers for the pricing constraints (denoted by the vector λ in equation (40) in the text) for the least squares optimization problem. A distinct multiplier is reported for each of the six securities listed in the first column. Numbers in parentheses are estimated standard errors. The standard errors are computed using the method described in Newey and West (1987) with $m = 15$. Descriptions of the securities and details of the data construction and sources are given in Section VI.A and Appendix B.

	$\beta = 1.00$		
Security	$\gamma = 0$	$\gamma = 5$	$\gamma = 15$
1	8.60	8.61	8.63
	(3.89)	(3.88)	(3.88)
2	1.81	1.80	1.77
	(0.96)	(0.97)	(0.98)
3	4.74	4.79	4.89
	(1.80)	(1.81)	(1.82)
4	−13.46	−13.47	−13.50
	(3.71)	(3.72)	(3.72)
5	3.76	3.75	3.73
	(0.89)	(0.88)	(0.88)
6	−5.56	−5.58	−5.64
	(1.77)	(1.77)	(1.78)

Table 4: Paper Table II: Lagrange multipliers for power utility.

5.4 Tables IV–VI: observable linear factor models

Read:

- Table IV shows that linear factor models have smaller distance measures than the power utility models.
- The equal-weighted return factor has $\delta = 0.286$ and $\chi^2 = 30.1$ when coefficients minimize specification error.
- The value-weighted return factor has $\delta = 0.289$ and $\chi^2 = 33.5$.
- Consumption growth has $\delta = 0.325$ when minimizing specification error, similar to the utility models.
- When coefficients are chosen to minimize χ^2, the consumption-growth factor has $\chi^2 = 12.0$ but $\delta = 1.57$.
- Table V shows that the Lagrange multipliers for the factor models remain large for managed portfolios.
- Table VI shows high correlations among the adjustment payoffs $\tilde{p}$ across proxies: all reported correlations exceed 0.87.

Economic message: The factor-model evidence is the paper's strongest warning. A model can look better by a classical χ^2 criterion while being much farther from any valid SDF in economically meaningful distance.

Table IV

Specification Errors for Linear Factor Models

The stochastic discount factor is $\hat{\alpha}'f_{t+1}$ where f_{t+1} contains a constant and one of the variable factors listed in the first column of the table, and $\hat{\alpha}'$ is the estimated parameter vector. Details of the data construction and sources are given in Section VI.A and Appendix B. The column labeled δ gives the least squares distances between the family of proxies and the family of stochastic discount factors. The column χ^2 gives the chi-square test statistic for the null hypothesis that all pricing errors are zero. Estimated standard errors for δ are given in parentheses. The standard errors are computed using the method described in Newey and West (1987) with m = 15.

Variable Factor	δ	χ^2
Coefficients Estimated by Minimizing the Specification Error		
Equally-weighted return	0.286	30.1
	(0.054)	
Value-weighted return	0.289	33.5
	(0.052)	
Consumption growth	0.325	38.9
	(0.052)	
Coefficients Estimated by Minimizing the Value of the χ^2		
Equally-weighted return	0.286	29.3
Value-weighted return	0.290	31.5
Consumption growth	1.57	12.0

that the chi-square criterion rewards variability in the discount factor proxy

Table 5: Paper Table IV: specification errors for linear factor models.

Table V

Lagrange Multipliers for Factor Models

The stochastic discount factor is $\hat{\alpha}'f_{t+1}$ where f_{t+1} contains a constant and one of the variable factors listed in the first column of the table, and $\hat{\alpha}'$ is the estimated parameter vector. The numbers reported are the Lagrange multipliers for the pricing constraints (denoted by the vector λ in equation (40) in the text) for the least squares optimization problem. A distinct multiplier is reported for each of the six securities listed in the first column. Numbers in parentheses are estimated standard errors. The standard errors are computed using the method described in Newey and West (1987) with m = 15. Details of the data construction and sources are given in Section VI.A and Appendix B.

	Variable Factor			
Security	None	Consumption Growth	Equally-Weighted Return	Value-Weighted Return
1	8.80	8.93	8.64	6.90
	(4.63)	(4.60)	(4.71)	(4.71)
2	1.71	1.50	−0.05	1.41
	(1.10)	(1.03)	(0.80)	(1.01)
3	4.80	5.42	6.10	6.30
	(2.20)	(2.32)	(2.10)	(2.04)
4	−13.52	−13.70	−13.00	−12.80
	(3.94)	(4.10)	(3.74)	(3.70)
5	3.80	3.64	3.10	2.90
	(1.0)	(1.09)	(0.90)	(0.80)
6	−5.60	−5.90	4.90	−4.70
	(1.50)	(1.50)	(1.50)	(1.40)

Table VI

Table 6: Ppaer Table V: Lagrange multipliers for factor models.

Correlations of $\bar{p}$ Across Proxies

This table reports the correlations among the estimated $\bar{p}$s for the proxies implied by four different models with alternative variable factors, where $\bar{p}$ [given in equation (28)] is the smallest adjustment in the least squares sense required to make the stochastic discount factor price the given set of assets right. The model with the variable factor labeled "None" has a discount factor that is constant. Details of the data construction and sources are given in Section VI.A and Appendix B.

Variable Factors	Consumption Growth	Equally-Weighted Return	Value-Weighted Return
None	0.99	0.87	0.88
Consumption growth		0.92	0.94
Equally-weighted return			0.96

factor models are quite high. (They all exceed 0.87). Abstracting from any

Table 7: Ppaer Table VI: correlations of the most-mispriced payoff adjustment across proxies.

6 Short summary

- The paper proposes distance-based measures for evaluating misspecified SDF models.
- δ is the distance from a proxy to the set of all SDFs that price the test assets.
- δ^+ is the distance from a proxy to the set of nonnegative SDFs that price the test assets.
- δ equals the maximum pricing error over unit-norm portfolios.
- δ^+ incorporates arbitrage-free pricing implications for derivative payoffs.
- The method is designed for model comparison when exact pricing is implausible.
- The empirical application uses six monthly returns from 1959 to 1990, including three managed portfolios.
- Power utility models have specification errors around 0.327–0.334.
- Adding a consumption externality barely changes the distance measure.
- Linear factor models based on equal-weighted and value-weighted returns reduce the distance to about 0.286–0.289.
- The reciprocal market return improves slightly over power utility, with distance about 0.311.

- δ and δ^+ are almost identical in the application, so positivity is not the binding issue.
- Classical χ^2 model comparison can be misleading because it rewards discount-factor variability.
- The paper's enduring lesson is to evaluate misspecified models by economic pricing errors, not only by exact-null hypothesis tests.

7 Conclusion

7.1 Core conclusion

Hansen and Jagannathan develop a way to compare stochastic discount factor proxies even when those proxies are known to be wrong. The correct comparison is not only whether the model is rejected by an exact pricing test, but how far the model's SDF proxy is from the family of SDFs that correctly price the relevant assets.

7.2 Economic intuition

A stochastic discount factor is a pricing rule. If a candidate pricing rule is wrong, its error should be measured by the worst pricing mistake it makes on the available payoff space, after normalizing for payoff size. This worst normalized error is exactly the least-squares distance from the proxy to the set of valid pricing kernels.

7.3 Takeaway

The paper gives asset pricing a practical model-comparison metric. Instead of ranking rejected models by how strongly they are rejected, rank them by the size of the adjustment needed to make their stochastic discount factors price assets correctly. That is why the Hansen–Jagannathan distance remains a central tool in empirical asset pricing.